4.7 Acid/base equilibria

Students will be assessed on their ability to:

a demonstrate an understanding that the theory about acidity developed in the 19th and 20th centuries from a substance with a sour taste to a substance which produces an excess of hydrogen ions in solution (Arrhenius theory) to the Brønsted-Lowry theory

b demonstrate an understanding that a Brønsted–Lowry acid is a proton donor and a base a proton acceptor and that acid-base equilibria involve transfer of protons

c demonstrate an understanding of the Brønsted–Lowry theory of acid-base behaviour, and use it to identify conjugate acid-base pairs

d define the terms pH, K_a and K_w, pK_a and pK_w, and be able to carry out calculations relating the pH of strong acids and bases to their concentrations in mol dm^{-3}

e demonstrate an understanding that weak acids and bases are only slightly dissociated in aqueous solution, and apply the equilibrium law to deduce the expressions for the equilibrium constants K_a and K_w

f analyse the results obtained from the following experiments:

- i measuring the pH of a variety of substances, e.g. equimolar solutions of strong and weak acids, strong and weak bases and salts
- ii comparing the pH of a strong acid and a weak acid after dilution 10, 100 and 1000 times

g analyse and evaluate the results obtained from experiments to determine K_a for a weak acid by measuring the pH of a solution containing a known mass of acid, and discuss the assumptions made in this calculation

h calculate the pH of a solution of a weak acid based on data for concentration and K_a, and discuss the assumptions made in this calculation

i measure the pH change during titrations and draw titration curves using different combinations of strong and weak monobasic acids and bases

j use data about indicators, together with titration curves, to select a suitable indicator and the use of titrations in analysis k explain the action of buffer solutions and carry out calculations on the pH of buffer solutions, e.g. making buffer solutions and comparing the effect of adding acid or alkali on the pH of the buffer

l use titration curves to show the buffer action and to determine K_a from the pH at the point where half the acid is neutralised

m explain the importance of buffer solutions in biological environments, e.g. buffers in cells and in blood (H_2CO_3/HCO_3^-) and in foods to prevent deterioration due to pH change (caused by bacterial or fungal activity).